



Ansys Fluent Simulation Report

Analyst	MONISH
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System Information

Application	Fluent
Settings	2d, pressure-based, inviscid
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:48:28 EDT
CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
OS	Windows

Geometry and Mesh

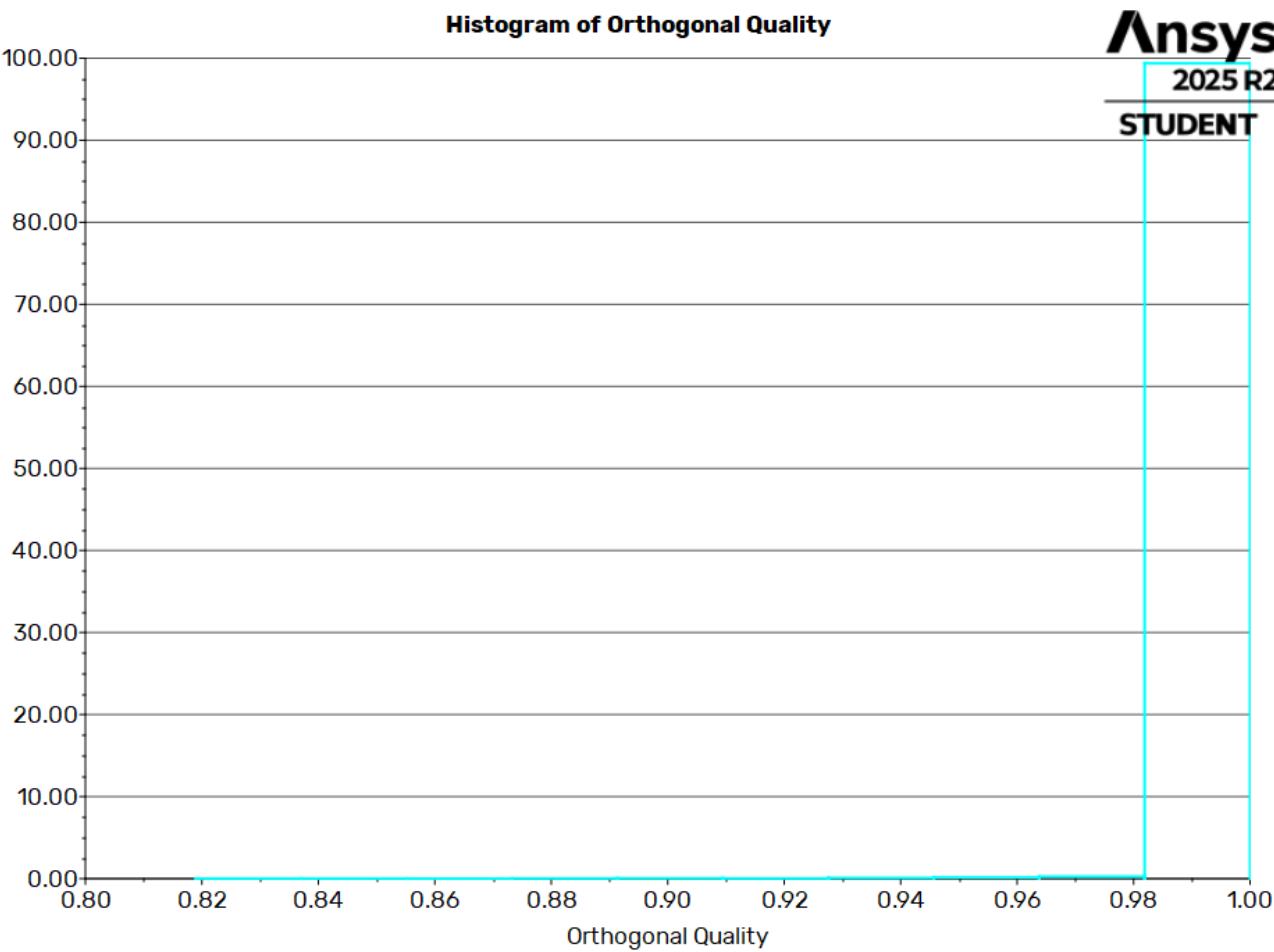
Mesh Size

Cells	Faces	Nodes
5744	11684	5940

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
square_block_model	Mixed Cell	0.81894743	2.9080553

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Steady
Viscous	Inviscid
Heat Transfer	Enabled

Material Properties

– Fluid	
– air	
Density	ideal gas
Cp (Specific Heat)	1006.43 J/(kg K)
Molecular Weight	28.966 kg/kmol
– Solid	
aluminum	

Cell Zone Conditions

– Fluid	
– square_block_model	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Porous zone?	no

Boundary Conditions

– Inlet	
– pressure_outlet	
Gauge Pressure [Pa]	498113.7
Mach Number	0.6
Temperature [K]	417
Component of Flow Direction (x,y)	(1, 0)
– pressure_inlet	
Gauge Pressure [Pa]	202650
Mach Number	2.5
Temperature [K]	288
Component of Flow Direction (x,y)	(1, 0)
– Wall	
boundary	wall
wall	wall

Reference Values

Area	1 m ²
Density	1.225 kg/m ³
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	square_block_model

Solver Settings

– Equations	
Flow	True
Energy	True

– Numerics	
Absolute Velocity Formulation	True
– Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Energy	0.75
Temperature	0.75
Explicit Momentum	0.5
Explicit Pressure	0.5
– Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
– Discretization Scheme	
Pressure	Second Order
Density	Second Order Upwind
Momentum	Second Order Upwind
Energy	Second Order Upwind
– Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

Number of Machines	1
Number of Cores	1
Case Read	19.526 seconds
Iteration	12.891 seconds
AMG	7.39 seconds
Virtual Current Memory	2.85717 GB
Virtual Peak Memory	2.86591 GB
Memory Per M Cell	386.514

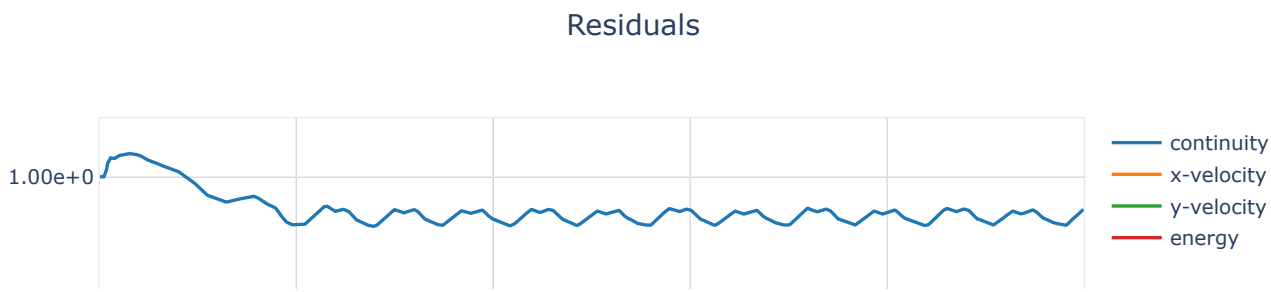
Solution Status

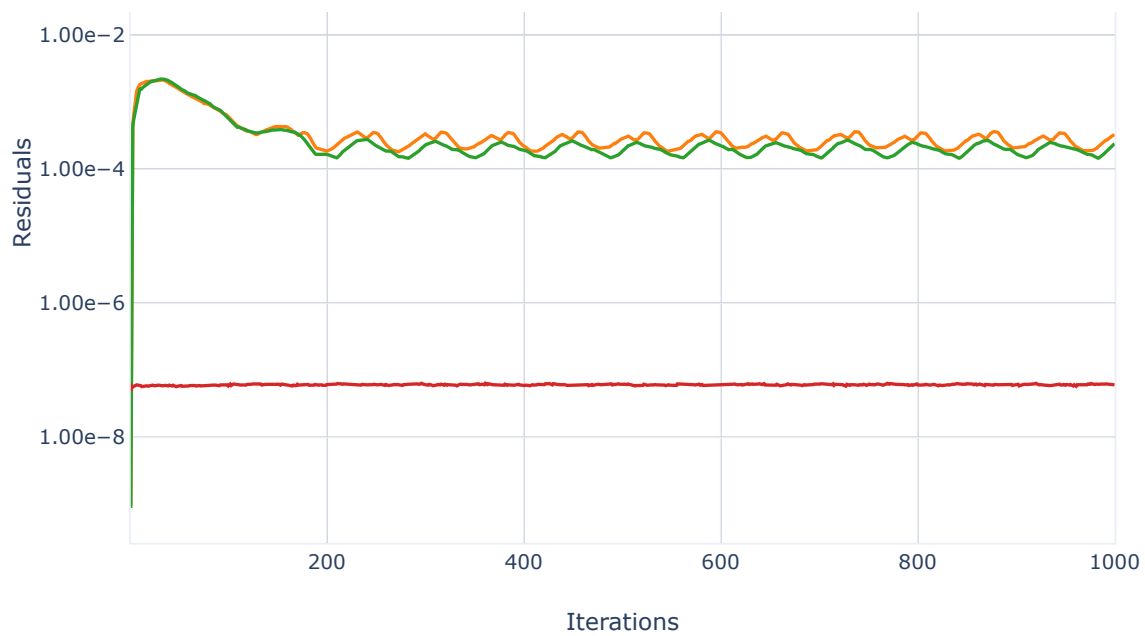
Iterations: 1000

	Value	Absolute Criteria	Convergence Status
continuity	0.3272029	0.001	Not Converged
x-velocity	0.0003273307	0.001	Converged
y-velocity	0.000240343	0.001	Converged
energy	6.068466e-08	1e-06	Converged

Plots

Residuals

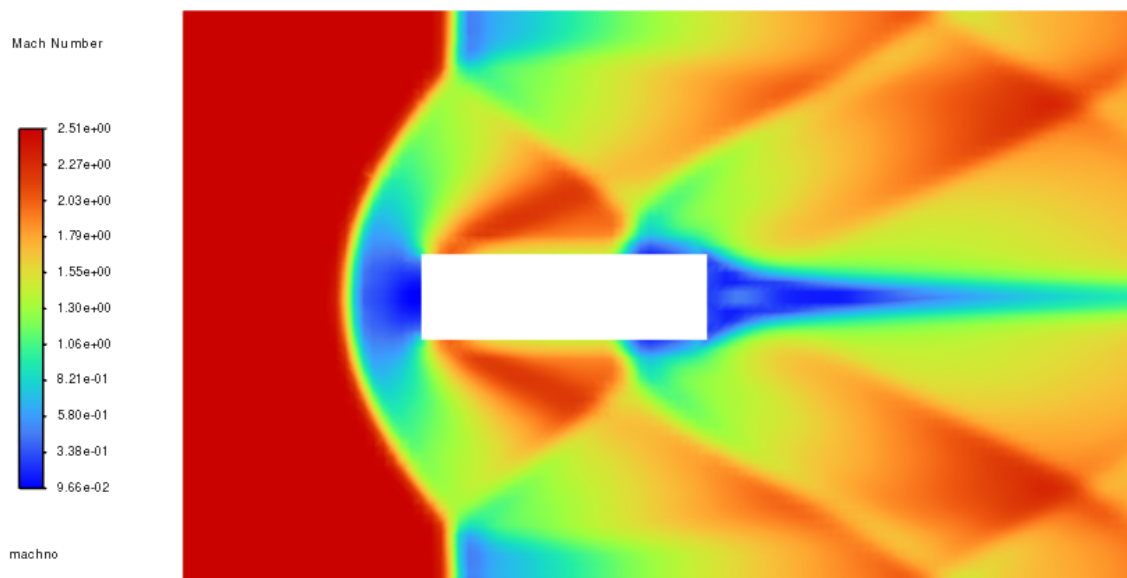




Contours

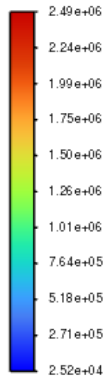
machno

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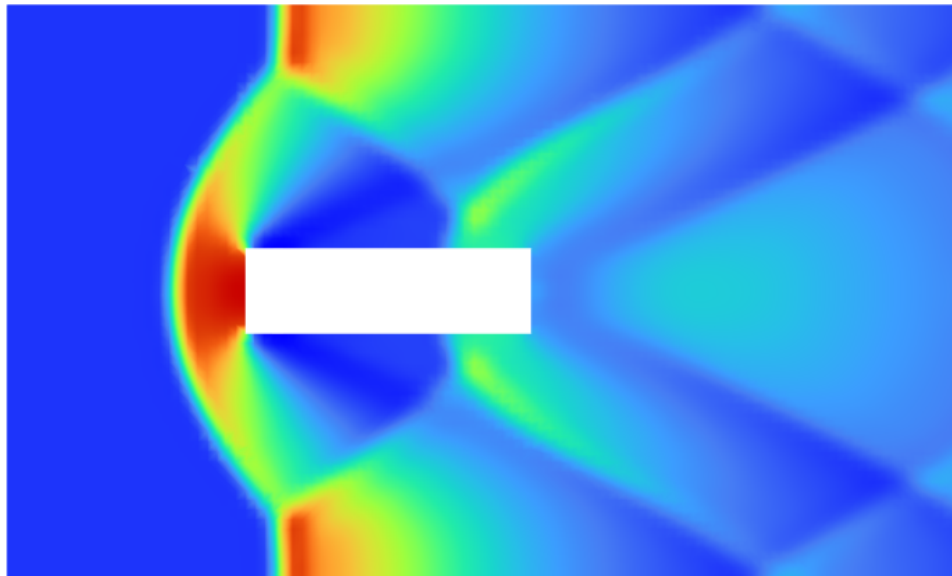


pressure-counter

Static Pressure
[Pa]

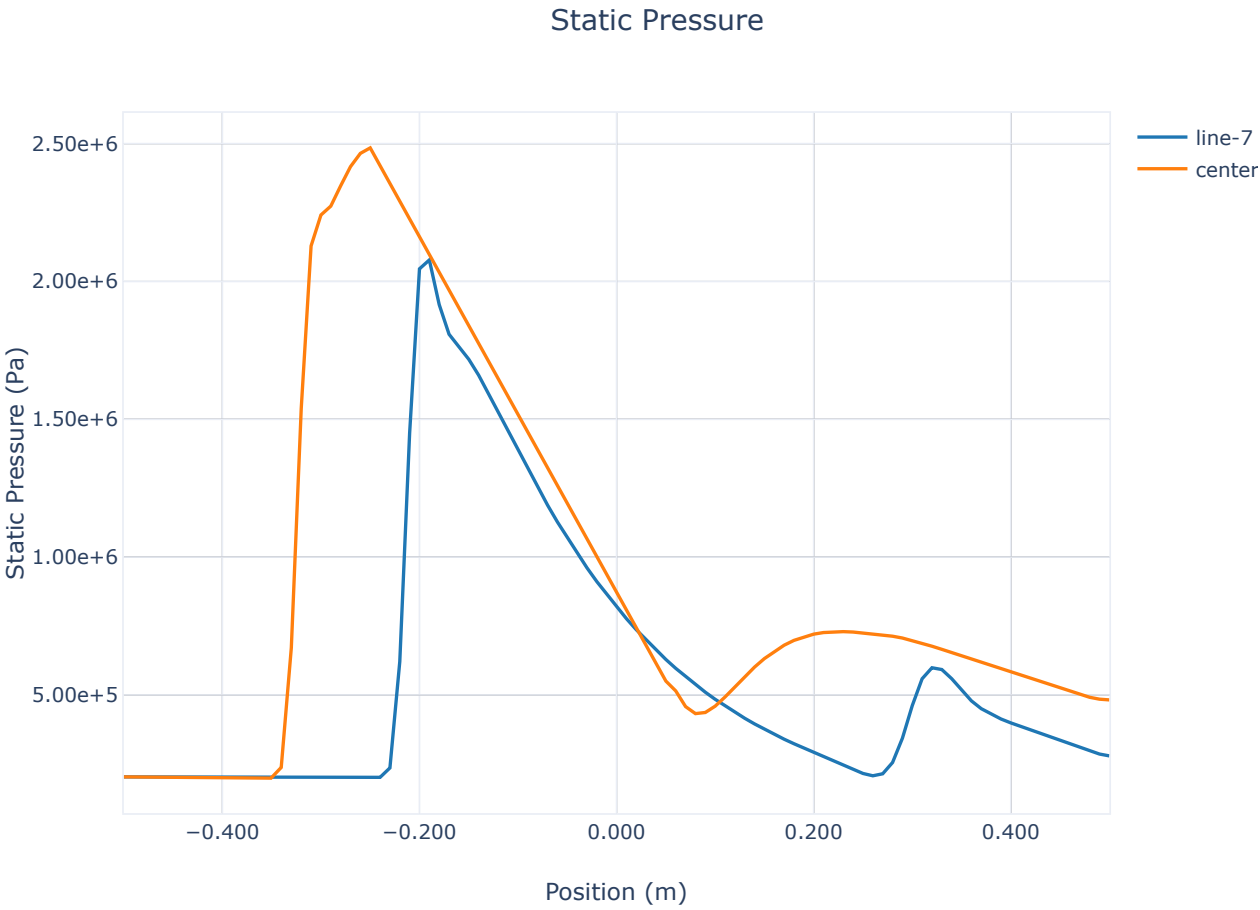


pressure-counter

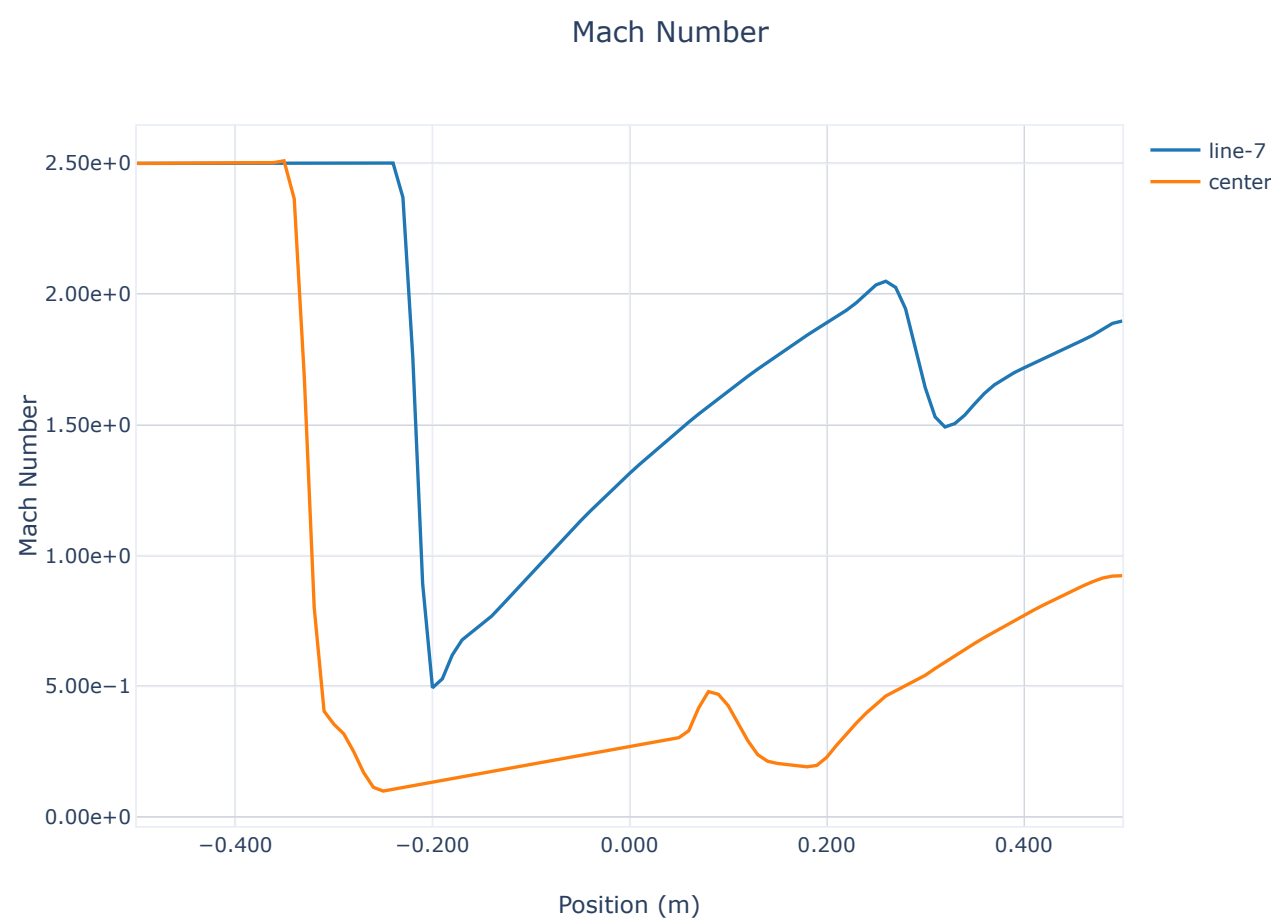


XY Plots

staticpressure

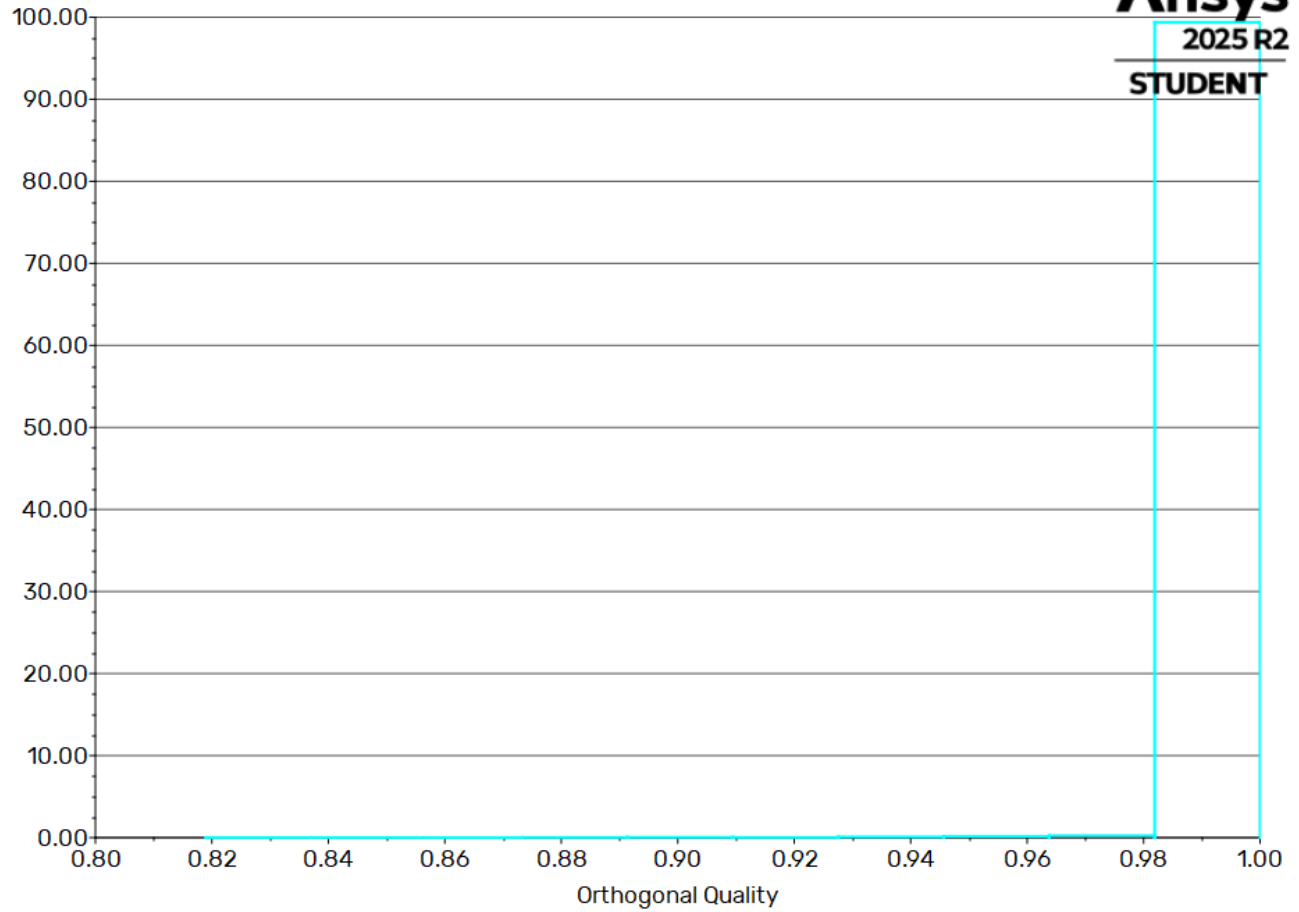


machno-plot

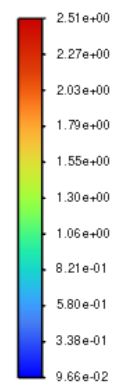


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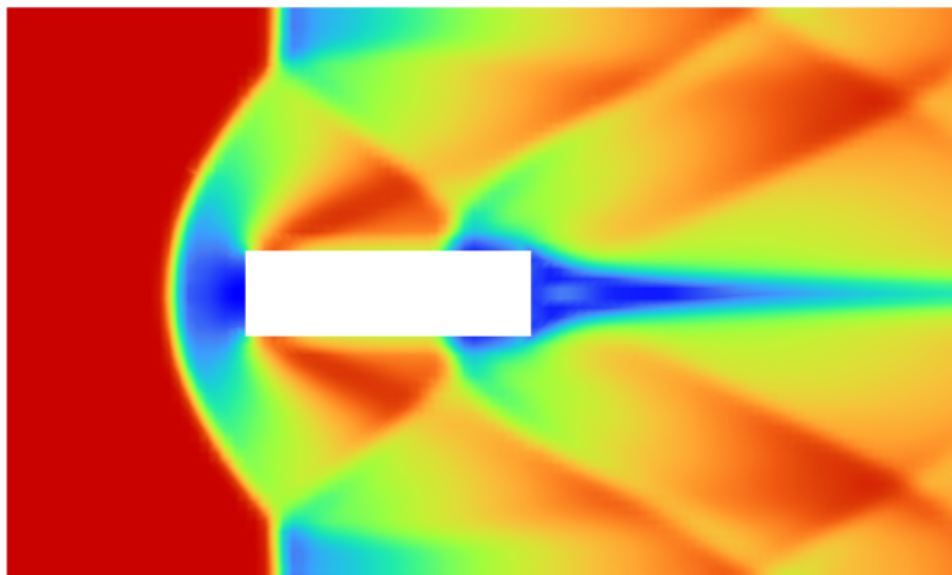
Histogram of Orthogonal Quality



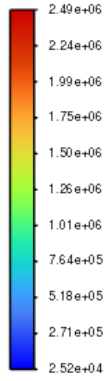
Mach Number



machno



Static Pressure
[Pa]



pressure-counter

